

# Distinctions as embodied experiences

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## *Abstract*

*Why and how individuals experience (or notice) something<sup>1</sup> is a relevant issue from many perspectives (psychological, philosophical, biological, etc). Even from an economic perspective this phenomenon (i.e. the emergence of distinctions) is at the core of decision-making processes in consumers, producers, and agents in general. This paper uses an evolutionary perspective in order to offer a plausible explanation — a model — for how distinctions emerge, showing that the phenomenon in question can be a self-recursive second order cybernetic case of approaches bridging the gap between phenomenology and modern cognition, namely neurophenomenology, and in line with those linking semiotic philosophy to cybernetics, namely cybersemiotics. Particularly, the model presented here could be classified under the realm of evolutionary neurophenomenological cybersemiotics, which refers to the scrutiny of human experience through modern cognition and Darwinian evolution using a cybernetic understanding of Peircean semiotics.*

## 1. Introduction

This discussion starts by acknowledging the limits of language, as any and all human creations, for describing phenomena<sup>2</sup> around, inside and/or among us. To overcome this obstacle, let's propose the development of an emergent language as the discussion unfolds. This is of course what most scientific contributions should do, if they want to be considered as such. However, the importance of mentioning it here refers to the need to make it a conscious act during the reading of this paper so that additional conclusions can be drawn at the end of the exercise. Presuming an agreement, let us consider the issue at hand, namely *How do we individuals experience things in order to construct our 'worlds' and how do distinctions emerge in or from these 'worlds'?* The definition of what an

individual's 'world' is, or what it represents, and how it is constructed is precisely the subject of this study. This paradoxical event should help the reader notice the restrictions of one of our most precious creations: language.

Please write a line or two about what you are experiencing right now:

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For the sake of the argument, we certainly hope that you, dear reader, did perform the task of trying to pinpoint the source(s) of experience(s); distinguished what it was (or what they were); and even tried describing it (or them) using words. Words are considered here as symbolic representations of your experience, and even though you probably knew what the experience was or what it felt like, you, we hope, had trouble finding the right way to describe it in a symbolic way. This is indeed a very hard thing to accomplish, and the remainder of this paper will try to show how evolutionary imputations to this eminently neurophenomenological process can help us describe how it could have taken place. We are by no means trying to patronize anyone here; this is just a simple exercise for illustrating the emergence process presented in this study.

Additionally, it is important to understand that we cannot undertake the analysis of the lines you just wrote, before developing a common emergent language; therefore, please accept a temporary apology and bare with the discussion for a moment until we reach an adequate point for returning to this issue. And please do not forget to try to make conscious the fact that we are engaged in the common endeavor of having a language emerge. Again presuming agreement, this paper will continue by situating itself in a scientific context (neurophenomenology), enabling a novel cybernetic understanding of Peircean semiotics and thereby the development of an alternative framework presented in section 3, from which our model of emergence emerges along with evolutionary neurophenomenological cybersemiotics.<sup>3</sup> Some conclusions and additional considerations are presented in section 5.

## **2. Context: Neurophenomenology**

The process in question (i.e. the reconstruction of our 'worlds') is emergent because its possible states as well as its relational constraints (transitional rules) can fundamentally change implying 'a new organization

and a new behavior, sometimes even a new identity' (Heylighen 1991: 89). This simply means that our 'worlds' and how we continuously reconstruct them can change without notice, without us being willing, without a fundamental reason or motivation, even without respect for relational constraints; in other words, the process is itself, and about, *in-principle undecidable questions* (Von Foerster 1992) rendering it at the core of ethics.

This does not mean, however, that the question about how we individuals experience things in order to construct our 'worlds', which in this paper is regarded as a question about how distinctions emerge in order to make up our 'worlds', cannot be undertaken or at least be subjected to documented speculation. In fact, regarding this process as an emergent second order cybernetic one already gives us hints about how to approach it.

For one thing, the scientific study of human experience is called phenomenology and within phenomenology many paths or orientations can be taken. Varela (1999) splits them up into three: *neuro-reductionism* or *eliminativism*, where a pessimistic surrendering of the experiencer takes place in the name neurobiological processes, much like in the ethological approaches of Hinde (1970) and Thorpe (1979); *functionalistic* orientations, where a dualistic concession gives rise, among others, to neural Darwinism (Edelman 1989 and 1992) and genetic and/or adaptive algorithms (Dawid 1996 and Holland 1975), which imply the identification of modular items of cognitive capacities and then putting them together through a theoretical framework (ranging from Phylogenetic to Ontogenetic adaptation as in Pallbo 1999) so that their unity amounts to an account of experience;<sup>4</sup> and *first-person* accounts of the irreducible nature of experience, where phenomenology rediscovers the primacy of human experience. Varela (1999) places his neurophenomenology in this category along with the cognitive semantics of Lakoff and Johnson (1999) as well as Searle's (1992) ontological irreducibility. Since we use neurophenomenology for modeling the emergence of distinctions, then we also place our *evolutionary neurophenomenological cybersemiotic* model under the first-person accounts of experience.

This means that in this study an experience will be anything that *happens to an experiencer*, so that any distinction potentially emerging from this experience could relate to a feeling, a perception, a sign, an element of cognitive structuration (i.e. the modular items of neural Darwinism), a relation, a combination of distinctions inducing a new experience, anything that only the experiencer could experience, distinguish, describe, and use. Therefore, here an individual's 'world' actually refers to our own 'inner world(s).'

*Neurophenomenology*

Let us now briefly explain Varela's (1999) description of the circulation between a first-person and an external account of human experience: first there is a gesture of *reflection*, which transforms experience into a sudden transient suspension of belief about what is happening, a doubt seeking to be resolved by identifying the source of experience; afterward there is an immediacy of experience surrounded by a diversity of horizons that lets us gain intimacy with our experience, an *intuitive* gesture, an act of opening up new possibilities within our habitual mind stream requiring multiple possibilities of the phenomenon as it appears, which are put to the service of whatever is our focus, and together give rise to an 'Aha!' experience that carries a force of conviction and lets us focus on the source of experience; a third moment requires this newly acquired gain in intuitive evidence to be described or translated into communicable items; Varela names these descriptions as *invariants* because it is variation that provides the broad conditions under which an experience is communicable, which represents an act of embodiment that *incarnates* what we experience; finally, training and steady learning that Varela highlights as needed to 'cultivate the skill to *stabilize* and deepen one's capacity for attentive bracketing and intuition, as well as skill for illuminating descriptions' (Varela 1999: 191).

Notice that at each and every moment Varela uses feelings (i.e. doubt, conviction, incarnation, stability) and human capacities (i.e. reflection, intuition, communication, description, training, learning) among others in his description, which means that feelings and capacities seem to be present throughout the whole process. This resembles in many ways the embodiment of distinctions process that Reyes and Zarama (1998) use for reconstructing the process of learning, on which we base our Tridimensional Recursive Learning Model — TRM (Parra and Yano 2002) and of which, we contend, Varela's neurophenomenological method is a special case, or vice versa, meaning that it is our model which could be a special case of Varela's.

In any case, first let us explain why we believe Varela's account of first-person experience can be viewed as a special self-recursive case of our second order cybernetic TRM model, and then point out the evolutionary dynamics underlining Varela's neurophenomenology.

*Neurophenomenology as self-recursive second order cybernetic learning*

Our TRM model involves a break in someone's ontological security that makes him or her declare ignorance, reminiscent of Varela's transient sus-

pension of belief generating doubt; then a distinction dissolving the break is offered, relating to the ‘Aha!’ experience that lets us focus on the source of experience; now the distinction introduced is put in context by grounding it in the individual’s history, linking directly to the description process that lets us incarnate an experience; finally, the distinction is embodied by putting it repeatedly into practice, which connects to Varela’s proposal of training in order to develop skill. The TRM also requires cyclic continuity for its recursive properties, and two agents (an instructor and a learner) running simultaneous and interchangeable learning cycles. We believe that Varela’s neurophenomenology represents the case of a single individual (one TRM agent): *Experiencing*, self-generating breaks; *Distinguishing*, self-offering distinction; *Grounding*, providing self-descriptions; and *Embodying* through self-training. These four stages constitute the EDGE model presented in section 4. Virtually the only difference between the two models (neurophenomenology and TRM) is that Varela regards embodiment as the incarnation of an experience through description (we call this grounding), while we use embodiment for denoting skills developed through training (i.e. repeated experiencing).

#### *Evolutionary dynamics of neurophenomenology*

Let us now bring up the implicit evolutionary dynamics typifying Varela’s neurophenomenological method, by keeping in mind that all social applications of Darwinian evolutionary models must include three main features: variation, selection, and inheritance (Hodgson 2002). In this paper, the word inheritance is replaced by the word retention, which accomplishes the same task, namely retaining structures or patterns that are adequate according to certain criteria (i.e. fitness for survival in the case of species). Varela’s neurophenomenology is an evolutionary process insofar as once an experience takes places, it is through variation (i.e. the surrounding diversity of horizons, or the multiple possibilities of the phenomenon as it appears) and selection (i.e. the focus toward the source of experience) that we manage to actually describe (i.e. incarnate) the phenomenon in question. Retention then appears through training, which lets us stabilize experiences and develop skills. We believe that retention should also take place earlier in the process based on the suitability of our experiential description, that is, the description’s adequacy in terms of certain criteria (i.e. the autopoietic effectiveness of an emergent structural coupling — Maturana and Varela 1998 — which obviously includes fitness for survival as mentioned before). This is basically why we believe that there is an underlying evolutionary process in progress during human

experiencing; however, we do think that it could be exploited further in terms of criteria for experiencing, distinguishing, grounding, and embodying experiences.

Evolution in neurophenomenology has to take place much faster than in the context in which it was discovered (or noticed), namely the evolution of species. However, this evolutionary process cannot be considered to take place instantaneously. If you had trouble trying to pinpoint, distinguish, and describe your experience symbolically (i.e. using words) at the beginning of this paper, then you can testify to this fact as well. In fact, there is a minimal time required for the emergence of neural events relating to cognitive events (Varela 1995a) and more so in processes also involving physiological dispositions, genetic predispositions, and heavy emotional loads, among others. According to Varela, ‘temporality is inseparable from all experience [...] this non-compressible time framework can be analyzed as the *manifestation* of the long-range neural integration in the brain linked to a widespread synchrony’ (1999: 193). Meaning not only that this process of synchrony is in part experiential, but more importantly, that there is a *dynamical reconstruction of invariants* (interpretants, as we shall designate them later) *continuously in progress* (Varela 1999, 1995a, and 1995b). Therefore, it is our contention that we can model the process defining and characterizing our ‘inner worlds’ using this neurophenomenological approach, Darwinian evolution, and an alternative cybernetic understanding of Peircean semiotics.

### *Recent approaches*

Brier (2003a) and Bateira (2003) have pointed out the ‘ingredients’ required for an adequate account of human knowledge — and of experience, insofar as ‘knowledge necessarily emerges from our lived experience’ (Varela 1999: 188) — namely modern cognition, phenomenology (Peircean semiotics), second order cybernetics, and information science. Brier (2003b) proposes a cybersemiotic model while Bateira (2003) points toward a naturalist paradigm. In terms of ingredients, our proposal is very similar to both Brier’s and Bateira’s; moreover, we add on to the critiques exposed by Breir (1992) in terms of mechanistic and reductionist approaches to knowledge and information. However, there are important differences that should be pointed out.

One of the main differences between neurophenomenology and cybersemiotics is that the former is concerned with experiences in general, while the latter mainly focuses on perceptions. Additionally, cybersemiotics (as typified by Brier 2003b) presents a ‘leveled’ usage of Peirce’s

interpretants that differs from the ‘staged’ usage discussed in the next sections. Nevertheless, since the *evolutionary neurophenomenological* process (out of which distinctions are to emerge) also uses a cybernetic understanding — in terms of distinctions — of Peirce’s interpretants, it should be placed under the *cybersemiotics* rubric as well. This in turn leads to the basic difference between cybersemiotics’ partial inclusion of evolution (i.e. to view mind as the ultimate product of evolution) and neurophenomenology’s fundamental use of it as previously noted. Cybersemiotics and the naturalist paradigm both acknowledge evolutionary contributions to their approaches, but they fail to explicitly include the variation, selection, and retention processes necessary for any social application of Darwinian evolution.

So far, we have clarified the reasons for defining the process in question as emergent, and have presented Varela’s neurophenomenological proposal, and its typification as self-recursive second order cybernetics’ learning, as well as its evolutionary dynamics. We have also illuminated some similarities and differences with recent knowledge (Bateria 2003) and information (Brier 2003b) approaches that point to the same mixture of theoretical distinctions for the scrutiny of human experience. In other words, we defined the context — neurophenomenology — from which a new cybernetic understanding of Peircean semiotics is developed next, so as to enable the design of a model reconstructing the emergence of distinctions from embodied experiences using an evolutionary neurophenomenological cybersemiotic framework.

### **3. Framework: Alternative cybernetic understanding of Peircean semiotics**

As mentioned before, we agree with Brier’s (2003a) ‘ingredients’ for a cybersemiotic conception; however, we will present an *alternative* perspective using basically these same ingredients, namely second order cybernetics, modern cognition (or cognitive biology), and Peircean semiotics in general (partially alluding to Sebeok’s 1976, 1989 biosemiotics). However, we decided to add two more ingredients, namely evolution as a process requiring variation, selection, and retention, and welfare economics (particularly human development).

Thus, we will undertake a cybernetic understanding of Peirce’s interpretants, evidencing the need for an evolutionary nature in this understanding, so as to redefine Firstness, Secondness, and Thirdness. Again, based on this understanding, we will then make additional evolutionary contributions to the neurophenomenological context presented in section

2 in order to delineate our EDGE model of emergence as well as evolutionary neurophenomenological cybersemiotics in section 4.

### *Cybernetic definition of interpretants*

An interpretant is defined here as ‘a modification of consciousness’ (CP 5.485) using the ‘analogous’ sense initially ascribed by Peirce. And at the same time, interpretants are also deemed as experiences, insofar as the smallest modification of consciousness is an experience itself (Lalor 1997). However, interpretants, besides being an experience, are here comprised of alternate bundles of embodied experiences (distinctions), presenting a definition reminiscent of Sen’s capabilities (1993) that implies alternate modifications of consciousness continuously reconstructing themselves through experience. Therefore, *interpretants are experiences, composed of embodied experiences (distinctions), which when experienced can be embodied as new distinctions, which will in turn potentially constitute a brand new interpretant*. Because once an interpretant is either used successfully<sup>5</sup> or considered adequate (in terms of any criteria or influencing factor including that of chance) it will no longer represent a mere experience, it is embodied as a newly available distinction.

This definition goes beyond the discussion of whether Peirce’s various presentations of his interpretants have semantic uniformity or distinctness (see Hardwick 1977, Short 1981, Lalor 1997) because when interpretants are regarded as experiences (and thereby as potential distinctions) composed of embodied experiences (distinctions), a single interpretant can at the same time incorporate immediate, dynamic, or final (if in fact such an ideal thing exists or happens) interpretants in the form of previously embodied or completely new experiences. We shall soon provide an example of how this could take place.

Moreover, our concept of interpretants is especially useful from a *human development* perspective (UNDP 1990) when capabilities are comprised of alternate bundles of choices or functionings (an individual’s beings and doings; see UNDP 1990 and Sen 1993) and these choices then, standing for embodied distinctions (Parra and Yano 2002), stem from distinctions that in turn stand for embodied experiences, which is in fact the focus of the current line of argument. Brier (2003a) considers interpretants as signs in our minds that make us recognize something as a thing and relates them directly to Von Foerster’s eigenvalues of cognition. Comparing it with our definition of interpretants, one could find them similar, but they differ not just in terms of the welfarist paradigm that we intend to explicitly promote, but also in terms of the eminently evolu-



tionary way in which interpretants are put together, discussed in section 4. Let us now consider why interpretants ought to be evolutionary in nature.

*Evolutionary cybernetic interpretants*

Pallbo (1997) justifies the use of evolutionary processes in the context of knowledge acquisition, which is directly related to our aim of ‘inner world’ creation, by showing how it helps overcome Meno’s dilemma, namely the fact that we cannot learn (notice, in this case) what we do not already know (experience) because we are unable to search for it or to recognize it should we stumble upon it. When knowledge is understood in evolutionary terms, it is allowed to grow and expand; it becomes ‘animated’ and can therefore evolve by itself. That is why we regard evolutionary contributions as indispensable in our framework and also why our definition of interpretants falls back on itself.

The process of constituting interpretants varies depending on the individual, and the specific time and place the individual is involved in, and can therefore take place emotively (based on feelings), cognitively (based on autopoietical relations or structural couplings), semiotically (based on signs), ethically (based on values or principles), historically (based on past experiences), physically (based on physiological dispositions or needs), genetically (based on biological predispositions), culturally (based on community and/or family customs or beliefs), reasonably<sup>6</sup> (based on logic or ‘common sense’), magically (based on superstitions, auguries, or cabbalas), or intuitively and/or creatively (based on any combination of these and other influencing factors that are probably yet unknown). All these factors are also experiences that, depending on the individual’s level of incarnation, are available as distinctions that influence the constitutions of interpretants, which taken together allude to those aspects attributing particular ‘meanings’ to experiences ‘partly for the sake of survival, partly for the sake of human flourishing beyond mere survival, and partly by chance’ (Lakoff and Johnson 1999: 91). Additionally, these factors directly relate to the ‘levels of existence’ characterizing Emmeche et al.’s (2003) emergence ontology, as well as Brier’s (2003b) informational theory, namely a quantum level with entangled causality; a physical level with efficient causality; an informational-chemical level with formal causality; a biological-semiotic level with non-conscious final causality; and finally a social-linguistic level with conscious goal-oriented causality. However, in this paper, these influencing factors are not placed in hierarchical levels of causality, but instead are considered to *interact with each other through a ‘widespread synchrony’* defined in Varela’s

(1999 and 1995a) work, bringing up yet another difference with mainstream cybersemiotics.

Therefore, *interpretants are experiences, made of embodied experiences (distinctions), resulting from the interaction of influencing factors (themselves new experiences or embodied distinctions depending on the degree of internalization of the experiencer) through a widespread synchrony, which can potentially become available distinctions for both, constituting an interpretant and/or influencing the constitution interpretants.*

Now, the hierarchical levels of causality proposed by mainstream cybersemiotics for us represent an unnecessary and restrictive relational constraint that hampers interactions between influencing factors. This means that we need to provide a new cybernetic version of Firstness, Secondness, and Thirdness, so that our alternative cybernetic understanding of Peircean semiotics can indeed have all influencing factors interacting with each other under widespread synchrony. In order to accomplish this, we shall start undertaking the analysis of the two lines you wrote at the beginning of this paper.

#### *Immediate interpretants (Firstness)*

Usually considered as a simpler sign, this is an interpretant of a preconscious complex of sensations (Lalor 1997), also chance, spontaneity, pure feeling (Brier 2003b), that does not specify a perspective. Our interpretation of this kind of interpretant is simply that of *undistinguishable experiences*. For example, let's consider hypothetically that when you tried to pinpoint the source of experience during the exercise at the beginning of this paper, you could not quite do it, much less distinguish it from others, and therefore did not manage to describe it; this would have amounted to an *undistinguishable experience* that yielded an *undistinguishable distinction*. Undistinguishable experiences (and distinctions) also encompass those unknown factors interacting with other influencing factors during the constitution of interpretants that are *hidden* even from us as experiencers and that we are therefore completely unaware of (i.e. experiences that are sub-, pre-, or unconscious).

#### *Dynamic interpretants (Secondness)*

This is the brute mechanical act that implies any exchange of energy, or the willpower of mind (Brier 2003b), or any further effect produced by the immediate effect involving a mental or muscular effort (Lalor 1997). Our interpretation of this kind of interpretant is that of *distinguished*

*experiences* that can be further described and embodied as *distinctions* or not, depending on the individual, its interpretants, and thereby its criteria (themselves distinctions in an interpretant) for putting interpretants together. Because if an individual manages to pinpoint the source of experience and to vaguely distinguish it, so that he is not quite able to describe it (i.e. readers that knew what the experience was, and what it felt like, but did not find the right way to describe it using words, again experiencing the shortcomings of language), then this could no longer be considered an *undistinguishable experience*, or Firstness. We would have already stepped into Secondness.

Similarly, for individuals clearly distinguishing an experience, but who out of freewill (or any other fixed interpretant or habit, that could include the lack of descriptonal capacities) did not describe it, and did not put it into practice either, so that it could become a fully incorporable experience to experiencers themselves and to others (i.e. a distinguishable distinction) resulting in distinguished experiences that are nevertheless undistinguishable distinctions. An example of this would be readers who halfway performed the exercise at the beginning of the paper, that is, they noticed and clearly distinguished an experience but disregarded the need to describe it. These examples are all taking place on the borderline between Firstness and Secondness, where the mere act of vaguely distinguishing already places us in Secondness without really taking us out of Firstness, and therefore clearly exemplify how an interpretant, as an experience, can be constituted by immediate and dynamic interpretants (experiences) at the same time, vis-a-vis distinguished experiences coexisting with undistinguishable distinctions in one bundle.

### *Final interpretants (Thirdness)*

This is defined as ‘the one interpretive result to which every interpreter is destined to come if the sign is sufficiently considered’ (Hardwick 1977, cited by Lalor 1997: 37) meaning that the human version of the final interpretant is an ideal. Therefore, our interpretation of this final interpretant is that of *platonic experiences*, which would, if they could be embodied and put into practice, yield *platonic distinctions* that should in turn take the symbolic form of *the most perfect account of an experience that words could convey*. For example, this means that you, dear reader, should have managed to do this in the two lines provided at the beginning of this paper, had you sufficiently considered your experience.

Brier (2003b) conceives of these as stable patterns or regularities or habits, at the last three levels of causality: formal causation (i.e. laws

of matter) in the chemical sciences; non-conscious final causation in the biological-semiotic arena; and goal-oriented final causation in social-linguistic contexts. It is probably because of his explicit need 'to respect the nature of the various subject areas and their specific types of knowledge that we have developed within the sciences and humanities' (2003b: 89). However, in so doing he is not only disregarding Peirce's warning in terms of regularities resulting from acquired habits of the mind, including the tendency to take habits themselves (see CP 6.268) in particular, the strong habit — fixed interpretant in our jargon — of recognizing regularities or stable patterns everywhere; but he is implicitly assuming that we have reached a threshold in scientific discovery as well. We are in no way proposing to disregard the contributions and advances within the sciences and humanities or to give up 'the intellectual hope that the sequence of interpretations [...] will ultimately represent reality' (Pape 1991, cited by Lalor 1997: 40). We are only suggesting to recognize these contributions and advances as stable patterns resulting from *our mind's overarching need to establish regularities and recognize stable patterns*, which should in fact be far from the interpretant with no further interpretation, or the *platonic experience* that if effectively embodied would lead to the *platonic distinction*.

Let us exemplify using the economic arena: the day all commodities (goods and services) are taken for what they 'really' are (i.e. their platonic distinctions) by all interacting agents (consumers and producers), we would be very close to the highly unlikely context of *perfect information*. It is easy to see why this is unlikely, let's consider a simple diamond ring: for the maker, this object is probably just another ring made of gold with certain specifications, following a specific design, holding a stone also of certain specifications, which together have a certain value; now for the future groom who wants to propose by giving it to his future fiancé, it probably represents more than what it does to the maker, and therefore the ring has a different value for him; considering what the ring could eventually mean for the bride who will get the ring, as well as its value for her, clearly shows how the existence of interpretants with no further interpretation would place limits on our humanity. Our finding would be similar if we were to look at different psychological and/or anthropological interpretations of the act of giving a present in order to propose marriage in different settings and/or cultures, only to mention a few possibilities.<sup>7</sup>

This links to the fact that the previously described cybernetic account of Peirce's interpretants should help shed light on the debate about whether Nelson and Winter's (1982) evolutionary economic interpretation of Polanyi's [1983 (1966)] tacit dimension of knowledge is in fact adequate (see Nightingale 2003, Johnson et al. 2002, Cowan et al. 2000,

Cohendet and Steinmueller 2000). *Tacit knowledge* would refer to *undistinguishable distinctions* encompassing undistinguished, ungrounded, and/or unincorporated experiences, while *codified knowledge* should be comprised of *distinguished, grounded, and embodied experiences as distinctions*. A detailed discussion about the tacit dimension of knowledge is beyond the scope of this paper; however, we want to point out that our cybernetic interpretants are to be placed at the core that debate.

We also regard Brier's claim that 'from an evolutionary view you will have to acknowledge that mind has emerged from nature' (2003b: 93) as reasonable, as long as it is not considered in the light of the assumption that 'living systems are the universe's way of becoming aware of itself' (2003b: 94) because, in our view, the whole cybersemiotic informational theory would then acquire an implicit anthropocentric overtone that is somewhat pretentious. On the other hand, it is our contention that there is in fact much more to the universe than we humans can experience (i.e. perceive, notice, evaluate, discover, and even dream of). Furthermore, it is troubling that evolution should keep on playing its traditional 'survival of the fittest' role only, when following Pallbo (1997) we have regarded evolution as an indispensable social application in Hodgson's (2002) terms, namely as any and all processes involving variation, selection, and retention. Therefore, in order to incorporate its contributions in our model meaningfully, let us include four evolutionary aspects to the list of factors influencing the constitution of interpretants. It should be clear that these evolutionary influencing factors are playing the role of experiences (distinctions or fixed interpretants depending on the individual) aiding the constitution of interpretants; therefore, we will refer to them as such.

We cannot overemphasize the importance of the fact that the previous cybernetic definition of interpretants along with Darwinian evolution are used not only for defining an evolutionary neurophenomenological cybersemiotic process, but also to typify 'sub-processes' acting at every single stage of it. This means that in order to effectively consider the *constitution of interpretants as evolutionary*, a process providing variation, selection, and retention of distinctions for comprising *experiencing, distinguishing, grounding, and embodying interpretants* should help individuals notice, generate, and evaluate variation, for the fixing of experiences as distinctions.

#### 4. The EDGE model of 'inner worlds'

In this model, interpretants (as well as the evolutionary process providing variation, selection, and retention for constituting these interpretants) are

present all throughout the evolutionary neurophenomenological cyber-semiotic process out of which distinctions emerge.

*Experiencing experiences: Noticing experiences*

This is the first stage of the emergence process out of which distinctions emerge and experiences are embodied. When an individual experiences an experience, interpretants are constituted with the basic motivational aim of noticing; therefore, interpretants active during the *experiencing* stage are the product of a process providing variation, selection, and retention of distinctions that help an individual notice.<sup>8</sup> As we mentioned before, this process usually leads the individual to doubt and generates a break in the individual's ontological security. In other words, when an individual is trying to notice something, a break is generated by *experiencing interpretants* that have been retained out of a few selected ones that were in fact variations generated with the aim of noticing. For example, if an individual is hungry, it is unlikely, though not impossible, that the interpretant constituted (as well as those prevalent during the constitution of the interpretant) for the act of noticing possible food corresponds to the one for noticing shelter, and vice versa, if the individual is cold.

*Distinguishing experiences: Generating variation*

Now, once this *experiencing interpretant* is constituted and active, it should eventually be resolved by providing the individual with various sources of experience. This should in turn activate the prevalent interpretants associated to processes providing variation, selection, and retention of distinctions that help the individual generate as much variation as possible, for the effective constitution of *distinguishing interpretants*.<sup>9</sup> This means that the generation of variation should be the main active motivation during the evolutionary process that eventually constitutes a *distinguishing interpretant*, but not just any variation; the generation of variation should be guided by the *experiencing interpretant* as well. For example, our hungry individual has an active *experiencing interpretant* for noticing food; this in turn yields to the activation of *distinguishing interpretants* that emerge out of an evolutionary process generating variation in terms of that prevalent *experiencing interpretant*.

*Grounding experiences: Evaluating variation*

Now, in order for us to be able to pinpoint the source of experience (through active *distinguishing interpretants*) and be able to describe it, we also need *grounding interpretants*<sup>10</sup> that should be the outcome of a simultaneous process providing variation, selection, and retention of selective distinctions that help the individual evaluate the provided variation, and in turn enable him or her to focus on a specific source of experience by describing it. Of course, the evaluation of variation, just as the generation of variation, should be directed by the *experiencing interpretants*' noticing aim. Therefore, out of the multiple possibilities (provided by *distinguishing interpretants*) of thousands of phenomena taking place around our hungry individual (noticed or not by *experiencing interpretants*), the *experiencing interpretant's* need to notice food should be the ubiquitous funnel of *grounding interpretants*. However, noticing food should not strictly be the only active or prevalent motivational aim at all times in *experiencing*, *distinguishing*, and *grounding interpretants*; it could be replaced by the need to stay alive, for example, or in fact any other aim, should the surroundings require the individual to do so (more on this ahead).

*Embodying experiences: Fixing distinctions in interpretants*

Finally *embodying or fixed interpretants* are those that, after a process providing variation, selection, and retention of distinctions helping individuals preserve suitable interpretants, allow individuals to fix these interpretants as prevalent during the constitution of interpretants. For example, the habit of recognizing regularities and making generalizations is a very strong *fixed interpretant* in humans. This means that our hungry individual can then, without being conscious, recognize what a possible source of food regularly does (i.e. time when fruits grow, or when fish come near shore, or deer go drink water), or where it could generally be found (i.e. on trees, in holes, etc.), or simply notice what other humans usually do (i.e. exchange commodities), or what the individual himself or herself often undergoes (i.e. getting hungry approximately every four hours). Particularly, this *fixed interpretant* — the habit of recognizing habits — is deeply rooted in us because it has been of great help throughout human history, especially for the development of science. However, these *fixed interpretants* also refer to pre-, sub-, or unconscious interpretants that individuals are not aware of. *Fixed interpretants* are in general very difficult to recognize or replace, and can actually hinder individuals from noticing and experiencing in the first place.

There are many reasons for having *fixed interpretants* that prevent us from experiencing, distinguishing, grounding, and embodying many experiences, so that they could become available distinctions. An individual's personality and/or culture are probably the most common *fixed interpretants*.

### *Evolutionary neuropsychological cybersemiotics*

*Evolutionary neuropsychological cybersemiotics* is the scrutiny of human experience through modern cognition and Darwinian evolution based on a cybernetic understanding of Peircean semiotics that is in line with, and promotes, *human development*. Through this process, individuals Experience something (a break occurs in the individual's ontological security triggered by **interpretants aiming to notice**), and then *Distinguish* the source of experience (separate it from other simultaneous experiences or disturbances — in systemic terms — by means of **interpretants that generate variation**), *Ground* what they are experiencing (describe it, define it, characterize it, exemplify it, associate it using 'inner world' terms and conditions in order to establish the experience's properties and structure, with the aid of **interpretants aiming to evaluate variation**), and when this new experience is used, *Embody* it as a distinction (i.e. it enters a bundle of distinctions constituting an interpretant by means of **interpretants aiming to fix**).

Even though at each stage of the EDGE model, there is a 'motivational' aim through which simultaneous evolutionary processes constitute experiencing, distinguishing, grounding, and embodying interpretants for noticing, generating, evaluating, and fixing experiences, respectively, it does not mean that these motivations should be the only aims at work. In other words, interpretants are constituted at each stage with a motivational aim in place; nevertheless, there are also other aims configuring interpretants that can become prevalent at any stage regardless of the task at hand (should there be one at all) and that can end up attributing a particular 'meaning' to the experience in question again for the sake of survival, the sake of human flourishing, and/or simply by chance, apart from all of the other strong *fixed interpretants* influencing the evolutionary constitution of interpretants that we are unaware of.

One might argue: why propose a model that can blow up at any time, and not be able to describe the process that is trying to typify? Well, we believe this to be an adequate way of handling emergent processes that refer to in-principle undecidable questions. Besides, if science is to become better at the task of 'representing reality' so that, in Peirce's terms,



the ultimate logical human interpretant corresponds to the final interpretant itself, then we all need to start relaxing our own relational constraints and those of our models. Of course, the model presented here could constitute an interpretant itself, probably a hopelessly misguided one, but nevertheless an interpretant. In fact, this paper's quest is to have the reader embody this experience as a new distinction that will hopefully become a prevalent interpretant during the subsequent constitution of interpretants. Should the reader have any misgivings about what the previous symbolic representations meant, we are already on our way to relaxing relational constraints, insofar as we have managed to generate a break!

In any case, it should be clear that *interpretants, whether as new experiences and/or fixed distinctions, are themselves in general involved in the widespread synchrony aiding the constitution of interpretants*, accounting for Varela's (1999) continuous dynamical reconstruction of invariants (aluding not just of *grounding interpretants* but to all of the others as well). Again, how interpretants and influencing factors for the constitution of interpretants interact is highly guided or misguided by chance. Paradoxically, because all influencing factors are themselves experiences that, depending on the individual and the situation, are internalized to different extents as available distinctions for making up interpretants, they can influence the constitution of interpretants. That is why new interpretants (i.e. spontaneous combinations of embodied experiences that 'animate' knowledge and allow it to evolve by itself) can also be regarded as new experiences that can potentially be turned into new distinctions if embodied.

Self-reference has always provided serious obstacles to mathematical and computational 'modelability.' However, in the case of evolutionary neurophenomenological cybersemiotics, as in neurophenomenology, we should not expect the list of interpretants, or that of factors influencing the constitution of interpretants as well as their alternative prevalence, to grow infinitely, but the number of widespread interactions grows exponentially every time an individual experiences something, anything, and a new distinction actually emerges. This is the case because an emergent distinction can potentially take part in any interpretant, or during the constitution of interpretants, and end up attributing a completely different meaning to an experience.

## **5. Discussion and conclusions**

Notice that we have replaced Brier's relational constraints in terms of levels of causality with stage relational constraints, mainly because he

sees interpretants as happening at different levels and we see them as happening at different moments, which means that both approaches instead of being alternative could be complementary or simultaneous. We believe this complementariness could emerge as long as human qualities and properties (i.e. mind) as well as human methods (i.e. stable patterns in the form of languages, models, etc.) created by our very own habits, are regarded as habits produced by *fixed interpretants* (i.e. that of recognizing patterns everywhere) instead of being regarded as the ultimate final interpretant. Our evolutionary neurophenomenological cybersemiotic EDGE model is, of course, trying to generalize a human habit (i.e. that of experiencing), and it is, at the same time, nothing more than a product of the human habit of recognizing regularities and making generalizations.

Now, venturing to ask an open question (such as ‘please write a line or two about what you are experiencing right now’) and then attempting to frame it using any kind of device is more than a double-edged sword, a self-inflicted policy. Nevertheless, let’s try to make sense of your answer (if there was one) in light of the model presented. In other words, now that presumably an emergent language has emerged between us, we can speculate about your personal account of an experienced experience.

### *Language and model in practice*

The very first exercise in this ‘interactive’ paper was to be conscious about the development of an emergent language. If the reader has managed to reach this point in the discussion, it is probably because we can already use common words to speculate. Let us then undertake yet another partial analysis of your answer.

- If in your writings you expressed doubt or uncertainty about your experience(s), you were probably generating a break on your ontological security using *experiencing interpretants*.
- If you expressed many mixed feelings and observations about your experience, then you were providing variation in terms of *distinguishing interpretants* in order to better distinguish your experience.
- If you responded with a particular thought or perception that adequately (with respect to your very own standards) described your experience, then you managed to select a *grounding interpretant* that best established the properties of your experience.
- If you went through all of the previous stages, then that was the precise intention of the exercise.
- If your experience was skeptical and led you to disregard the exercise, for any valid reason (i.e. not being in the mood, not having the time,

deciding that it would not be worth it anyway, etc.), then probably a *fixed interpretant* is limiting you in ways that you yourself as an experiencer cannot notice, distinguish, or ground, and might in fact be preventing you from further experiencing.

- If your experience was completely different from the previously mentioned options, then perhaps this evolutionary neurophenomenological cybersemiotic account is indeed hopelessly misguided and we greatly apologize for attempting to deceive anyone.

However, notice that we just used a common emergent language to try to rediscover your own experience. For accomplishing this, you should have incarnated some of the experiences experienced while reading this paper as new distinctions (in order to effectively embody them, we suggest that you reread the paper). Now, if in the future you eventually use these new distinctions in Immediate, Dynamic, Experiencing, Distinguishing, Grounding, or Embodying interpretants, or maybe even use them to self-expose a fixed interpretant (habit) that was hampering your experiences, you are actually turning these distinctions into possibilities for acting (Roth 2002), and they will potentially become available choices (Parra and Yano 2002). And this is ‘good’ because the more interpretants, the more experiences an individual will be able to experience, distinguish, ground, and embody as distinctions available for more interpretants.

If you, dear reader, happened to experience, distinguish, ground, and (we hope) embody any new experiences, then this article effectively put into practice the process of emergence of distinctions. Particularly in the introduction, we tried generating a break by asking you to write what you were experiencing, then we offered a distinction that could dissolve the break through a new understanding of interpretants (section 3), afterward we exemplified how this could take place in terms of evolutionary neurophenomenological cybersemiotics (section 4), and in this last section we used an emergent language and the EDGE model to evidence the embodiment (or at least the partial incarnation) of experiences triggered by the break. In other words, we have structured our argument and its presentation (i.e. the *embodiment of experiences process for the emergence of distinctions*) as an autopoietic self-constructed *embodiment of distinctions process where choices emerge*.

## Notes

1. This ‘something’ literary refers to anything: objects, beliefs, connections, feelings, signs, thoughts, values, codes, reflections, concepts, perceptions, relationships, observations, definitions, etc.

2. Phenomena refers to anything we explore or scrutinize; therefore, these may or may not relate to us in significant or relevant ways.
3. We realize that this nomination is not very handy at all, but it nevertheless encompasses explicitly all the theoretical distinctions making up our model.
4. According to Brenner (1998), there are at least three problematic issues when trying to use evolutionary algorithms (i.e. adaptive or genetic) for modeling social evolution, namely they generally disregard the past history of individuals; the definition of fitness in social evolution is subjectively determined (on this, also see Sen 2002); and they fail to consider motivational effects during selection, while the 'selection process is influenced by the distribution over strategies in a complicated way which cannot be reproduced by common evolutionary algorithms' (Brenner 1998: 282).
5. 'Successful' means that the interpretant has managed to establish a structural coupling (Maturana and Varela 1998) between the individual and its environment, which in turn imposes 'some kind of limit that puts a "constraint" on the possible ways an organism can exist as an organism' (Brier 2003a: 108), basically alluding to the usual 'survival' role in which evolution is generally considered.
6. 'Rationally' could denote utility maximization in the traditional economic sense; instead, 'reasonably' refers to the use of reason for scrutiny.
7. There are even more dramatic examples in health economics: consider a patient going to the doctor with acute chest pain and willing to do or pay whatever in order to receive immediate medical attention, while the doctor, finally determining that the pain is only a minor side effect from a bad shoulder, considers eventually not to charge the patient at all.
8. This motivational aim could relate to inherited and/or innate psycho-physiological predispositions (Beer 1982) involved in most ethological approaches (e.g. the biological scrutiny of behavior, analyzed in detail by Brier 1992 using Bitterman 1965 and Lorenz 1976), but it is by no means exhausted there. So the difference between traditional ethological motivation and the motivational aims considered here is that the evolutionary process providing variation, selection, and retention of interpretants for *noticing* entails more than biological predispositions, as should be evident from the section on evolutionary cybernetic interpretants presented before (being in line with the concerns of Thorpe 1979, Hinde 1970, and Madsen 1974, also discussed by Brier 1992).
9. These are acting in place of Varela's (1999) *variations*.
10. These are acting in place of Varela's (1999) *invariants*.

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